

Lab 04 Instructions

Due Feb 14 5:00 PM PST

STAT240 Spring 2025

2025-02-07

To solve these problems please submit a pdf generated from a qmd or Rmd file, showing all code required to solve the problems. You can use the template `Lab_04.qmd`. For any text answers please type them directly (making them bold), do not use `cat` inside an R code chunk, as these do not render clearly, and often run off the page, meaning they can't be read.

Problem 1a)

Write an R function called `my_lm` that takes a vector `x` and a vector `y` and returns the slope `a` and y-intercept `b` of the line $f(x) = ax + b$ that minimizes the sum of squared errors between $f(x_i)$ and `y[i]`. Your function must not make use of the R functions `lm`, `lm.fit`, or any R function that does linear regression (you may, however, use functions like `sum` or `mean`). Your function should return a named list with two elements named `a` and `b`. Provide your code. You may use the following code as a template:

(2 points)

```
my_lm <- function(x, y) {  
  ...  
}
```

Problem 1b)

Verify that the function `my_lm` that you wrote in Problem 1a) is correct by running `my_lm` on a vector `x` of length three with values given by the first three digits of your student number (one value per digit), and a vector `y` given by the last three digits of your student number. Print the values that `my_lm` returns. Then, do the same regression with the R function `lm` and

print the resulting slope and y-intercept. Note whether or not the values returned by the two methods are the same.

(1 points)

Problem 2)

In the canvas folder for this lab is the SQLite database `xcoretail.sqlite` discussed in class. For each month in the range January 2011 to August 2011 (inclusive), calculate Z Co Retail's total revenue for that month. Provide code, and plot the revenue by month (your y-axis can be 1, 2, 3, ... instead of dates, but make sure to label the axes clearly with units).

A description of the database follows. This database has two tables: `sales` and `prices`. The table `sales` contains one row for each sale that the company made during the reporting period. The table `sales` has three columns:

- `StockCode`: a string indicating the product identifier.
- `Quantity`: an integer indicating how much product was sold.
- `InvoiceDate`: a string indicating the date of the sale.

The table `prices` is a table indicating the price of each product in the company's inventory and has four columns:

- `StockCode`: a string indicating the product identifier (matching the identifiers in the table `sales`).
- `Description`: a string describing the product.
- `UnitPrice`: the price in USD of the product.

While we want to eventually plot the total sales across all products, we will first do this for a single product.

Part a)

Using the class code as a template, construct a plot showing the total sales of the item with stock code 71053 for the 8 months from 2011-01 to 2011-08. To do this you will need to first extract the total number of items sold each month and then, using information in the `prices` table, compute the value of these sales, in USD. Plot this revenue against the month.

(3 points)

```
library(RSQLite)
library(DBI)
```

Part b)

Now, potentially using a similar approach to Part a, compute the total revenue per month across **all** items in this database.

(4 points)